

FoodPro Architecture, Workflow, and Roles

This reference summarizes the implemented architecture relationships, operational process flows, and built-in role/permission model used by the current FoodPro deployment.

1. ER-Style Relationship Overview

Core Master Data

- Site (Project) -> User, Inventory, Production, MaterialRequest, FoodWaste, PurchaseRequest, PurchaseOrder
- Ingredient -> Recipe.ingredients, Inventory, InventoryLot, InventoryTransaction, PurchaseRequestItem, PurchaseOrderItem, GoodsReceiptItem, SupplierPriceHistory, FoodWaste
- Recipe -> Production, MenuPlan, POS Recipe Mapping, FoodWaste

Production and Material Flow

- Production -> MaterialRequest via source_production_id
- Production -> InventoryTransaction on start/complete consumption posting
- Production -> FoodWaste via production_id and recipe_id
- MaterialRequest -> Procurement acknowledgement -> unlocks production start

Procurement Flow

- PurchaseRequest -> PurchaseRequestItems
- PurchaseRequest -> PurchaseOrder -> PurchaseOrderItems
- PurchaseOrder -> GoodsReceipt -> GoodsReceiptItems
- PurchaseOrder / GoodsReceipt -> SupplierInvoice
- PurchaseOrderItem -> SupplierPriceHistory

POS and Forecasting

- POS Source -> POS Sales Orders -> POS Sales Items
- POS Recipe Mapping links POS items to FoodPro Recipe
- POS sales + Production + MenuPlan + Waste -> ForecastScenario / ForecastSnapshot analytics

2. Module Workflow Overview

End-to-End Production Workflow

- 1. Chef creates production request.
- 2. System snapshots required ingredients and creates linked draft material request.
- 3. Chef submits production request for approval.
- 4. Project Manager / Operations Manager reviews, approves, rejects, or requests changes.
- 5. If approved, material request status changes to pending procurement acknowledgement.
- 6. Procurement Officer acknowledges the material request.
- 7. Chef starts production after procurement acknowledgement.
- 8. Chef completes production and inventory/cost posting is performed.

Food Waste and Cost Intelligence Workflow

- 1. Waste is posted by ingredient, recipe, batch, or location.

- 2. Estimated waste cost is calculated from ingredient or production costing.
- 3. High-value waste requires approval.
- 4. Historical waste is used to drive recipe-level reduction guidance in production planning.
- 5. Food cost reporting aggregates production cost by date, location, meal type, and recipe.

Location and Visibility Workflow

- 1. Company -> Region -> Location -> Kitchen -> Store/Warehouse hierarchy is maintained in Site.
- 2. Users are assigned by site_id and allowed_site_ids.
- 3. visibility_scope controls whether a user sees assigned locations only or subtree scope.
- 4. Server-side locationScope filtering prevents access to non-assigned project data.

3. System Role Catalog

- Administrator: Full system access across all modules and locations
- Operations Manager: Cross-functional control of projects, inventory, production, procurement, waste, reports
- Project Manager: Project-level production review, approval, waste approval, material request visibility
- Chef: Production creation, recipe/menu control, material request creation, production start/complete, waste posting
- Storekeeper: Inventory and transfer operations
- Procurement Officer: Procurement execution, supplier handling, MR acknowledgement
- Production Supervisor: Production execution and control
- Quality Controller: Quality and waste compliance
- Finance Controller: ERP, exports, reporting, forecasting visibility
- Custom Role: RoleProfile-based permission set configured in User & Role Management

4. Role-Permission Matrix

Permission / Action	Chef	Project Manager	Storekeeper	Procurement Officer	Admin
Create Production Request	Yes	Optional	No	No	Yes
Edit Draft Production Request	Yes	No	No	No	Yes
Submit Production Request	Yes	No	No	No	Yes
Review Production Request	No	Yes	No	No	Yes
Approve / Reject / Request Changes	No	Yes	No	No	Yes
Create Material Request	Yes	No	No	No	Yes
Acknowledge Material Request	No	No	No	Yes	Yes
Start Production	Yes, after MR ack	Optional	No	No	Yes
Complete Production	Yes	Optional	No	No	Yes
Post Food Waste	Yes	Yes	Optional	No	Yes
Approve High-Value Waste	Role-configurable	Yes	No	No	Yes
Manage Inventory / Transfer	Optional	Yes	Yes	No	Yes

Manage Procurement / Suppliers	No	Optional	No	Yes	Yes
Manage Users / Roles	No	No	No	No	Yes

5. Key Design Notes

- Relational tables are used for auth, procurement, and POS integration; entity_records provides extensible JSON-backed models for many operational modules.
- Server-side location scoping protects project-level visibility and prevents users from seeing unassigned project names and data.
- Custom roles are supported through RoleProfile while still inheriting the core access_level model of admin / manager / user.
- Production and material request approval flow is enforced by permission checks and workflow status transitions in the backend.